



I2C – LPS25 테스트

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Mango-M32F2 I2C-LPS25 대기압 sensor 테스트

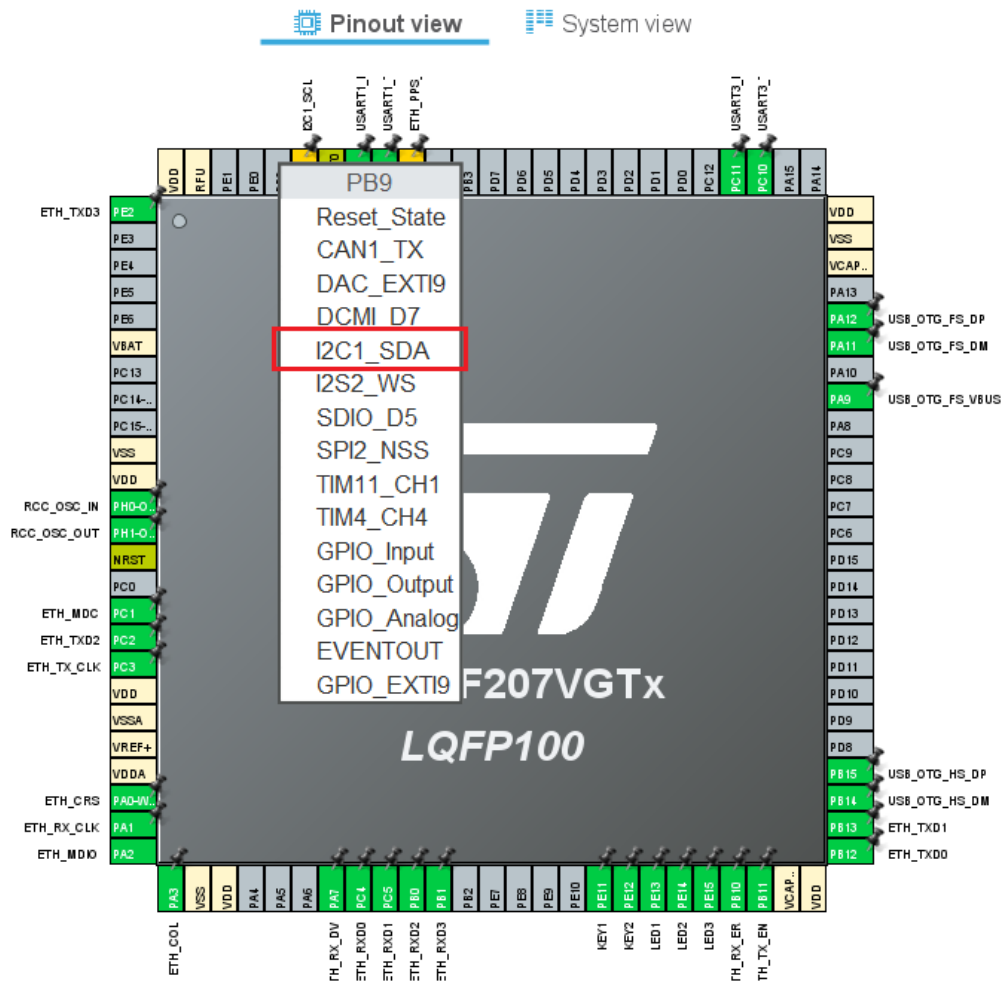
I2C pinmap

I2C – PB8, PB9

PB7	93	PB7	UART1_TX
PB8	95	PB8	I2C1_SCL
PB9	96	PB9	I2C1_SDA
PB10	47	PB10	MII_RX_FR

Pinout view

■ PB8-I2C1_SCL, PB9-I2C1_SDA 선택.



I2C 설정

- “Connectivity” 클릭, “I2C1” 클릭.
- “Mode” - “I2C” 클릭.

The screenshot displays the 'bma280_prj.ioc - Pinout & Configuration' window. The 'Pinout & Configuration' tab is active, showing a list of components on the left and configuration options on the right. In the left sidebar, under the 'Connectivity' category, 'I2C1' is selected and highlighted with a red box. The right pane shows the 'I2C1 Mode and Configuration' settings. The 'Mode' dropdown menu is set to 'I2C' and is also highlighted with a red box. Below the mode selection, there are tabs for 'Configuration', 'NVIC Settings', 'DMA Settings', 'GPIO Settings', 'Parameter Settings', and 'User Constants'. The 'Configuration' tab is active, showing a 'Reset Configuration' button and a section for 'Configure the below parameters :'. This section includes 'Master Features' with 'I2C Speed Mode' set to 'Standard Mode' and 'I2C Clock Speed (Hz)' set to '100000'. It also includes 'Slave Features' with 'Clock No Stretch Mode' set to 'Disabled' and 'Primary Address Length' set to '7 bit'.

LPS25 Test Project

Code 생성 후 main.c 수정

```
37 /* Private define ----- */
38 /* USER CODE BEGIN PD */
39
40 #define LPS25H_SLAVE_ADDRESS (0x5D << 1)
41
42 #define COUNT_AVERAGE_VAL 1000
43
44 /* USER CODE END PD */
45
46 /**
47  * @brief Set the slave address according to SA0 bit.
48  * @param SA0_Bit_Status: LPS25H_SA0_LOW or LPS25H_SA0_HIGH
49  * @retval None
50  */
51 void LPS25H_Init(void)
52 {
53     uint8_t dataVal = 0x00;
54     uint8_t dataValRead = 0x00;
55
56     //PowerOn
57     dataVal |= LPS25H_MODE_ACTIVE;
58     /* Output Data Rate selection */
59     dataVal |= LPS25H_ODR_25Hz; //Output Data Rate: P - 25Hz, T - 25H
60     /* Interrupt circuit selection */
61     dataVal |= LPS25H_DIFF_ENABLE; //interrupt circuit enabled
62     /* Block Data Update selection */
63     dataVal |= LPS25H_BDU_READ; //output registers not updated until MSB and LSB reading
64     /* Serial Interface Mode selection */
65     dataVal |= LPS25H_SPI_SIM_3W; //3-wire interface
66     SENSOR_LPS25H_Write_Bytes(LPS25H_SLAVE_ADDRESS, LPS25H_CTRL_REG1_ADDR, 1, &dataVal);
67     SENSOR_LPS25H_Read_Bytes(LPS25H_SLAVE_ADDRESS, LPS25H_CTRL_REG1_ADDR, 1, &dataValRead);
68     printf("LPS25H_CTRL_REG1_ADDR init read : 0x%02X\r\n", dataValRead);
69     dataVal = 0x00;
70     dataVal |= LPS25H_P_RES_AVG_512; //LPS25H Pressure resolution selection AVG_51
71     dataVal |= LPS25H_T_RES_AVG_64; //LPS25H Temperature resolution selection AVG_64
72     SENSOR_LPS25H_Write_Bytes(LPS25H_SLAVE_ADDRESS, LPS25H_RES_CONF_ADDR, 1, &dataVal);
73     SENSOR_LPS25H_Read_Bytes(LPS25H_SLAVE_ADDRESS, LPS25H_RES_CONF_ADDR, 1, &dataValRead);
74     printf("LPS25H_RES_CONF_ADDR init read : 0x%02X\r\n", dataValRead);
75 }
76 /* USER CODE END 0 */
```

LPS25 Test Project

```
222  /* USER CODE BEGIN 2 */
223  printf("I2C-LPS25 test\r\n");
224  httpd_init();
225
226  HAL_Delay(5000);
227  LPS25H_Init();
228  HAL_Delay(300);
229  SENSOR_LPS25H_Read_Bytes(LPS25H_SLAVE_ADDRESS, LPS25H_WHO_AM_I_ADDR, 1, dataVal);
230  printf("Device id : 0x%02X\r\n", dataVal[0]);
231
232  /* USER CODE END 2 */
```

LPS25 Test Project

```
273     calcData = (float)data32Val_sum / COUNT_AVERAGE_VAL / 4096.0;
274
275     printf("Pressure: %5.2f hPa\r\n", calcData);
276
277     data16Val=0;
278
279     /*
280      * @brief Read LPS25H output register, and calculate the raw temperature.
281      * @param int16_t *raw_data: temperature raw value.
282      * @retval LPS25H_ERROR or LPS25H_OK.
283      */
284     SENSOR_LPS25H_Read_Bytes(LPS25H_SLAVE_ADDRESS, LPS25H_TEMP_OUT_L_ADDR + 0x80, 2, buffer);
285
286     printf("buffer[0]: 0x%02X, buffer[1]: 0x%02X\r\n", buffer[0], buffer[1]);
287
288     data16Val = (((int16_t)buffer[1]) << 8) + (int16_t)buffer[0];
289
290     calcData = (((float)data16Val/480.0) + 42.5);
291
292     unsigned_data16Val=data16Val;
293     printf("data16Val: 0x%4X, Temperature: %f°C\r\n", unsigned_data16Val, calcData);
294 }
295 /* USER CODE END WHILE */
```

Test 결과 확인

- CS-LPS25 센서 보드를 8Pin Cable을 이용 CON7에 연결함.
- Device ID – 0xBD 정상적으로 읽힘.
- 대기압과 온도 정보가 정상적으로 읽힘.

VT COM14 - Tera Term VT

File Edit Setup Control Window Help

```
I2C-LPS25 test
LPS25H_CTRL_REG1_ADDR init read : 0xCD
LPS25H_RES_CONF_ADDR init read : 0x0F
Device id : 0xBD
Pressure: 1015.55 hPa
buffer[0]: 0x51, buffer[1]: 0xDE
data16Val: 0xDE51, Temperature: 24.535418° C
Pressure: 1015.95 hPa
buffer[0]: 0x57, buffer[1]: 0xDE
data16Val: 0xDE57, Temperature: 24.547916° C
```

Thank You

